

proxy_forward_http_challenge.sh

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.27 (Challenges), §11.4.69 (sink-side positive evidence), §11.4.1 (no false-FAIL), §11.4.68 (no fail-open), §11.4.2/§11.4.5 (captured evidence)

Overview

Anti-bluff Challenge that proves the **live HTTP forward proxy** (default `http://localhost:34128`) actually forwards real end-user traffic. It drives two real sub-probes through the proxy — a plain-HTTP GET to a 204 endpoint and a CONNECT-tunnelled HTTPS GET — and asserts the through-proxy `%{http_code}` matches the expected code, cross-checked against a **direct** fetch of the same URL. Response headers (including Squid's `Via:` line) are captured to an evidence file, so every PASS cites a real artefact.

Prerequisites

- The base proxy stack UP with the HTTP proxy listening on 34128.
- `curl`, `bash`, `awk`, `grep`; `tests/lib/evidence.sh` present (sourced).
- READ-ONLY: the script is a plain proxy client — it never starts, stops, restarts, or reconfigures any container.

Usage

```
bash challenges/scripts/proxy_forward_http_challenge.sh
# override target / evidence dir:
HTTP_PROXY_URL=http://localhost:34128 \
CHALLENGE_EVIDENCE_DIR=/path/to/evidence \
  bash challenges/scripts/proxy_forward_http_challenge.sh
```

Environment: `HTTP_PROXY_URL` (default `http://localhost:34128`), `HTTP_PROXY_PORT` (default 34128), `CHALLENGE_EVIDENCE_DIR` (default `qa-results/challenges/<run-ts>`), `CURL_MAX_TIME` (default 20).

Exit codes

Code	Meaning
0	PASS — ≥ 1 sub-probe proved forwarding and none FAILED.
1	FAIL — a real proxy defect (proxy missed but the same URL was reachable directly).
3	SKIP — honest non-applicable: no endpoint reachable (third-party/network outage, §11.4.3).

Anti-bluff decision (per sub-probe, via `proxy_conn_verdict`)

- proxy code in expected → **PASS**
- proxy miss + same URL reachable **directly** → **FAIL** (§11.4.68, no fail-open)
- proxy miss + direct miss → **SKIP** (§11.4.1, no false-FAIL on a site outage)

Aggregate: **FAIL** if any sub-probe FAILs; **PASS** if ≥ 1 PASSes and none FAIL; otherwise **SKIP**.

Edge cases

- A 204 endpoint that goes down SKIPs its sub-probe (not a proxy fault); the HTTPS sub-probe can still carry the run to PASS.
- The captured `Via: ... proxy-squid` header is hard client-side proof the bytes transited the proxy, beyond the HTTP status code alone.

Outputs

`<evidence-dir>/http/forward_http_evidence.txt` (per-probe codes, verdicts, captured response headers) plus per-probe raw header dumps.

Related scripts

- `proxy_socks5_challenge.sh` — the SOCKS5 sibling.
- `proxy_cache_challenge.sh` — Squid cache-HIT proof.
- `run_proxy_challenges.sh` — the bank runner.
- `tests/lib/evidence.sh` — the sourced anti-bluff helper library.

Last verified: 2026-07-01 (live run: PASS, Via: 1.1 proxy-squid (squid/6.13) captured).